

Tutorial 5

Problem 1

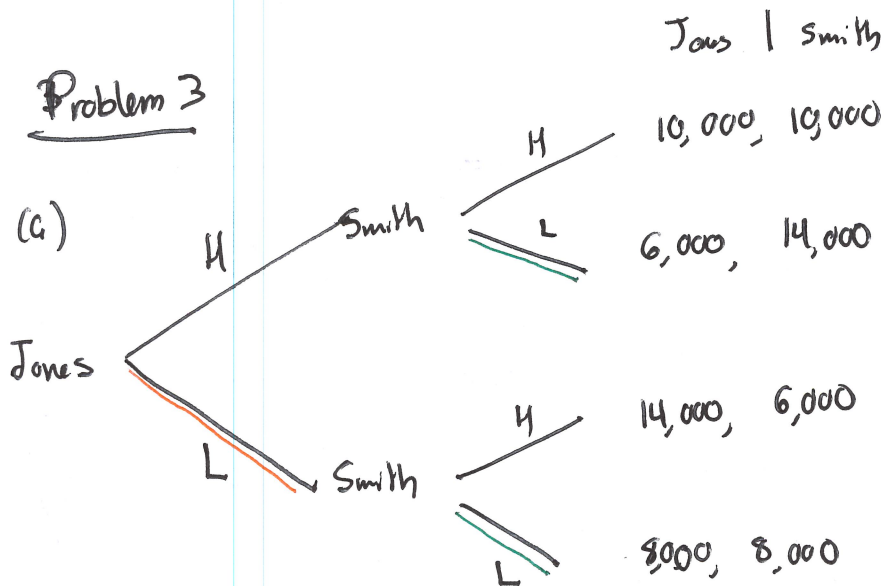
		Broadcast		
		H	M	L
Astanding	H	3, 5	5, 7	4, 9
	M	0, 4	8, 6	5, 2
	L	1, 2	0, 5	3, 3

- (a) Note that astanding has a dominant strategy, which is M
- (b) No, the best reply of broadcast changes with the strategy of Astanding
- (c) No, since Broadcast doesn't have a dominant strategy
- (d) The NE is (M, M) with a payoff outcome of (8, 6)

		Russia	
		NI	I
US	T	5, 4	7, 3
	MC	3, 5	6, 9

- (a) Best Response US (NI) = T
- (b) Best Response US (I) = T
- (c) Best Response Russia (T) = NI
- (d) Best Response Russia (MC) = I
- (e) The only NE is (T, NI) with a payoff of (5, 4)

Problem 3



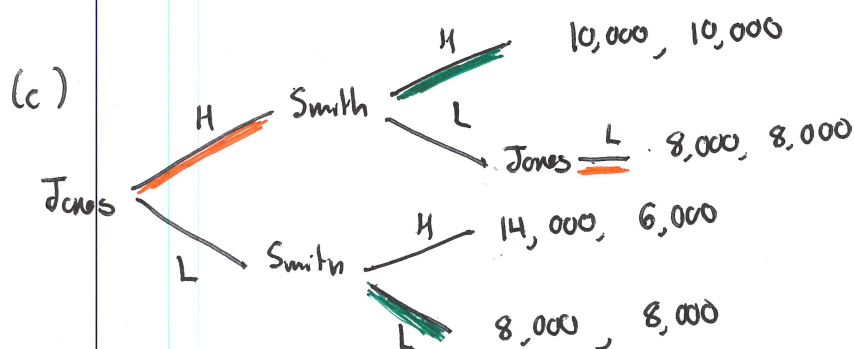
The equilibrium strategies are:

Jones: Low

Smith: Low when following H by Jones
Low when following L by Jones

$\Rightarrow$ The outcome is both end up playing L on equilibrium with each profits of 8,000

(b) No, because whenever it is Smith's turn to move he always wants to deviate to Low. Thus, it is not a credible promise.



The equilibrium strategies are:

Jones: H in first node, L in second node

Smith: $\begin{cases} H & \text{if Jones played H} \\ L & \text{if Jones played L} \end{cases}$

$\Rightarrow$ The outcome is that now Jones and Smith play High, with a payoff of 10,000 each

$\hookrightarrow$ Why? by committing to match price, Jones is able to make a credible promise that induces Smith to play High if ~~Smith~~ Jones chooses High first. By doing this both end up better

Problem 4

- (a) q_A = barrels of A
 q_B = barrels of B
 $Q = q_A + q_B$

(q_A, q_B)	Q	P	A' Profit	B' Profit
10, 10	20	125	1250	1250
20, 20	40	50	1000	1000
10, 20	30	75	750	1500
20, 10	30	75	1500	750

- (b) The payoff matrix is:

		B	
		10	20
A	10	1250, 1250	750, 1500
	20	1500, 750	1000, 1000

For both A & B it is a dominant strategy to produce 20
 so (20, 20) is a dominant strategy equilibrium.

- (c) Each country would earn 1250 as seen above.

- (d) ~~No~~ Yes, since they always have an incentive ~~to~~ to deviate to 20, since 20 is a dominant strategy.

Problem 5.

$$(a) \quad \left. \begin{array}{l} MR = 120 - 2Q \\ MC = 30 \end{array} \right\} \begin{array}{l} MR = MC \\ 120 - 2Q^M = 30 \\ Q^M = 45 \\ P^M = 120 - 45 = 75 \end{array}$$

(b) Each firm produces 22.5

$$\text{Total Profits are: } 45 \times 75 - 30(45) = 2025$$

$$\Rightarrow \text{Each firm earns: } 1012.5$$

(c) If both firms produce 30 $\Rightarrow Q = 60 \Rightarrow P = 60$

$$\text{Each firm profit is: } 30 \times 60 - 30 \times 30 = 900$$

(d) Suppose Firm 1 produces 22.5 $\Rightarrow Q = 52.5$

Firm 2 produces 30 $\Rightarrow P = 67.5$

$$\text{Firm 1 profits: } (67.5 - 30) 22.5 = 843.75$$

$$\text{Firm 2 profits: } (67.5 - 30) (30) = 1125$$

(e)

		Firm 2	
		22.5	30
Firm 1	22.5	1012.5, 1012.5	843.75, 1125
	30	1125, 843.75	900, 900

30 is a dominant strategy for both firms

so (30, 30) is a dominant strategy eq and thus also a Nash equilibrium.